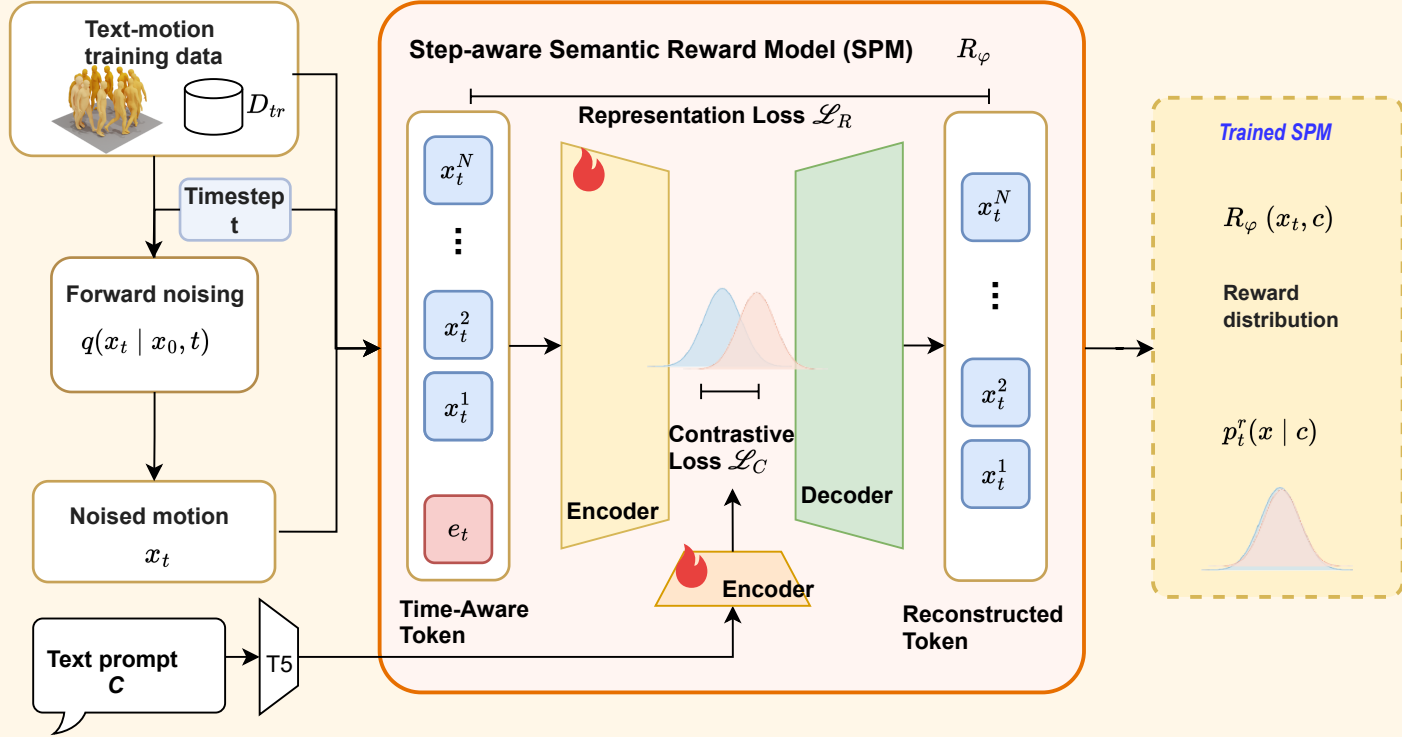
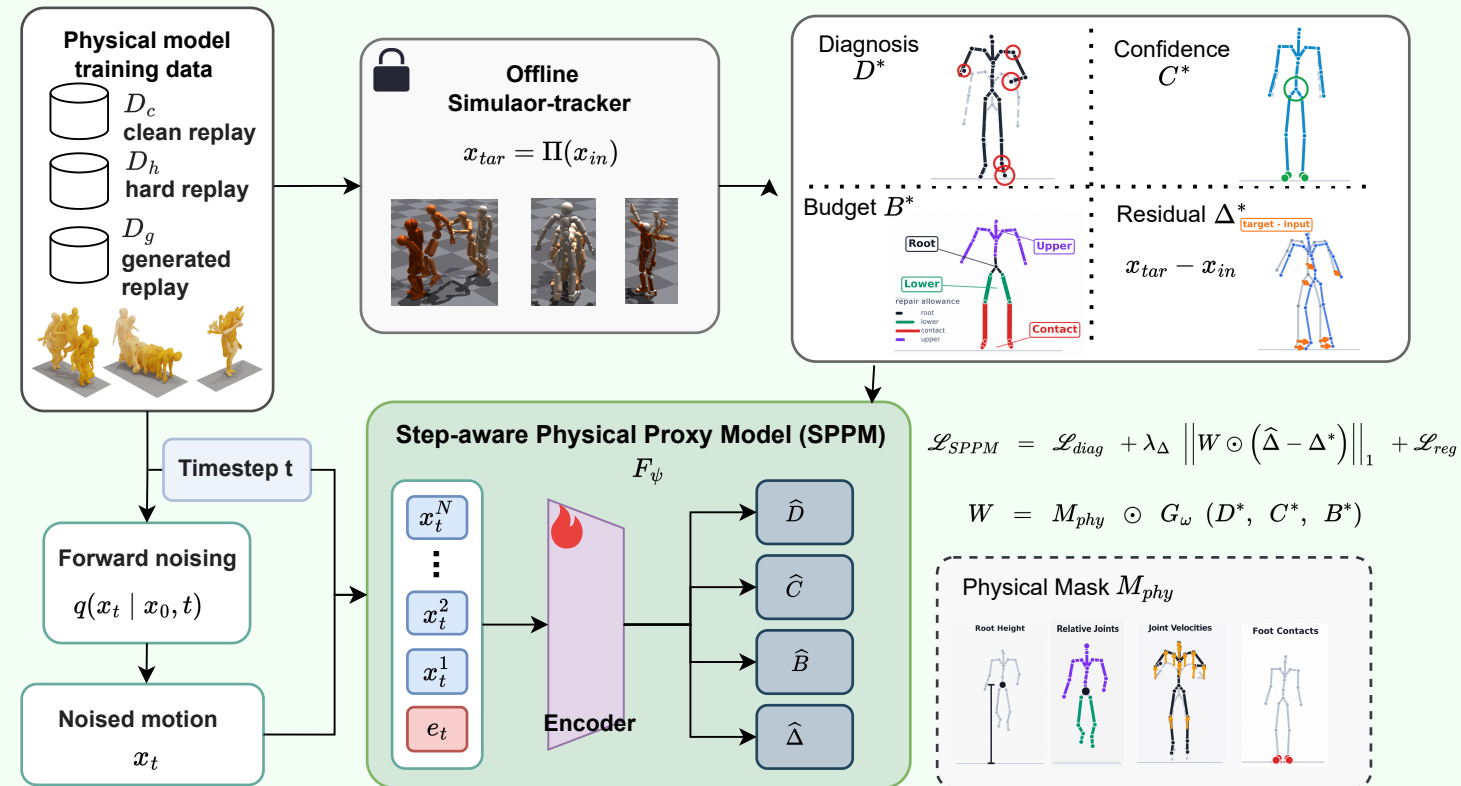


Stage 1: Train Step-aware Semantic Reward Model (SPM)



Stage 2: Build Offline Physical Data and Train Step-aware Physical Proxy Model (SPPM)



Stage 3: Inference-time Plug-and-Play Semantic-Physical Denoising

